

Kryptos – Background

Including Updated Information on Part 3

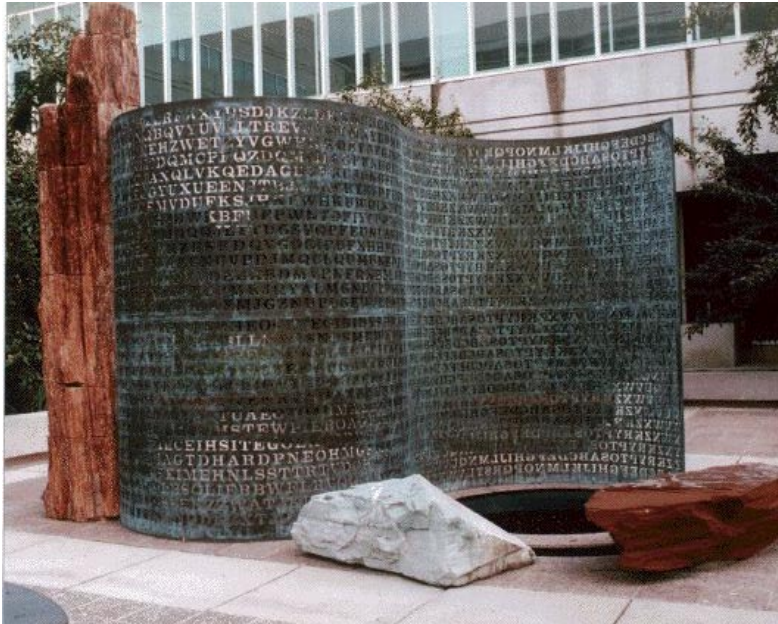
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Background

Credit for much of what is known about Kryptos must go to two people. As far as I am aware Douglas Gwyn is the first person to have publicly posted an accurate transcription of the text contained on the sculpture. Jim Gillogly was the first to post a solution to the first three parts. While someone within the CIA had solved the same three parts slightly earlier, this was not made public until after Mr. Gillogly announced his solution. Thus is it Jim that I “blame” for getting me, and probably most other members of this group, involved with Kryptos.

Kryptos is located in a courtyard at CIA headquarters in Langley, Virginia. It was commissioned by the CIA and created by artist Jim Sanborn with the help of Ed Scheidt, former head of the

CIA's Cryptographic Center. Dedicated in October 1990, the secret message it contained remained a mystery for nearly a decade, until the solution for the first three parts of the message was announced in 1999. The last part remains unsolved.

The inscriptions on the two panels (again thanks to Douglas Gwyn) are shown below, the message on the left, and the Vigenere table on the right. The blue bands separate the message into its four different parts.

EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGVHKKQETGFQJNCE GGWHKK?DQMCPPQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBQLBQCRBTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFFUPWNTDFIYCUQZERE EVLDKFEZMOQQJLTUGSYQPFUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXXHDKRF FHQNTGPUAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVBNFQXEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG ENDYAHROHNLSRHEOCPTIOIBIDYSHNAIA CHTNREYULDSLSSLNOHSNOSMRWXMNE TPRNGATIHNRRARPESLNNELEBLPIACAE WMTWNDITEENRAHCTENEUDRETNHAEOE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSPAMHHEWENATAMATEGYEERLB TEEFOASFOTUETUAEOTOARMAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB AECTDDHILCEIHSITEGOEASDDRYDLORIT RKLMLHAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSSTRTVDOHW?OBKR UOXOGHULBSOLI FBBWFLRVQQPRNGKSSO TWTQSJSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	ABCDEFGHIJKLMNOPQRSTUVWXYZABCD AKRYPTOSABCDEFGHIJLMNQUVWXXZKRYP BRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT CYPTOSABCDEFGHIJLMNQUVWXXZKRYPTO DPTOSABCDEFGHIJLMNQUVWXXZKRYPTOS ETOSABCDEFGHIJLMNQUVWXXZKRYPTOSA FOSABCDEFGHIJLMNQUVWXXZKRYPTOSAB GSABCDEFGHIJLMNQUVWXXZKRYPTOSABC HABCDEFGHIJLMNQUVWXXZKRYPTOSABCD IBCFGH IJLMNQUVWXXZKRYPTOSABCDE JCDEFGH IJLMNQUVWXXZKRYPTOSABCDEF KDEFGH IJLMNQUVWXXZKRYPTOSABCDEFG LEFGH IJLMNQUVWXXZKRYPTOSABCDEFGH MFGH IJLMNQUVWXXZKRYPTOSABCDEFGHI NGH IJLMNQUVWXXZKRYPTOSABCDEFGHIJ OH IJLMNQUVWXXZKRYPTOSABCDEFGHIJL PIJLMNQUVWXXZKRYPTOSABCDEFGHIJLM QJLMNQUVWXXZKRYPTOSABCDEFGHIJLMN RLMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ SMNQUVWXXZKRYPTOSABCDEFGHIJLMNQU TNQUVWXXZKRYPTOSABCDEFGHIJLMNQUV UQUVWXXZKRYPTOSABCDEFGHIJLMNQUVW VUVWXXZKRYPTOSABCDEFGHIJLMNQUVWX WVWXXZKRYPTOSABCDEFGHIJLMNQUVWXX XWXXZKRYPTOSABCDEFGHIJLMNQUVWXXZ YXZKRYPTOSABCDEFGHIJLMNQUVWXXZK ZZKRYPTOSABCDEFGHIJLMNQUVWXXZKRY ABCFGH IJLMNQUVWXXZKRY
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The solution to the first three parts of the message (*with spaces and punctuation added*) are shown below. Note that three words with apparent "errors" are highlighted in red:

Part 1 - Vigenere Cipher with key word PALIMPSEST

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF
ICLUSION(sic).

Part 2 - Vigenere Cipher with key word ABSCISSA

IT WAS TOTALLY INVISIBLE. HOW'S THAT POSSIBLE? THEY USED THE EARTH'S
MAGNETIC FIELD. X THE INFORMATION WAS GATHERED AND TRANSMITTED
UNDERGRUUND(sic) TO AN UNKNOWN LOCATION. X DOES LANGLEY KNOW ABOUT THIS?
THEY SHOULD. IT'S BURIED OUT THERE SOMEWHERE. X WHO KNOWS THE EXACT

LOCATION? ONLY WW. THIS WAS HIS LAST MESSAGE. X THIRTY-EIGHT DEGREES, FIFTY-SEVEN MINUTES, SIX POINT FIVE SECONDS NORTH. SEVENTY-SEVEN DEGREES, EIGHT MINUTES, FORTY-FOUR SECONDS WEST. ID BY ROWS.

Note that, on the right hand panel, the normal alphabets running across the top and bottom, as well as down the left side of panel, appear to be for decorative purposes only. The alphabets that actually serve to provide the coordinates used during the solution of Parts 1 and 2 are the alphabets starting with KRYPTOSABC that run across the second row and down the second column.

Part 3 - Transposition

SLOWLY, **DESPARATLY**(sic) SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED. WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN, WIDENING THE HOLE A LITTLE, I INSERTED THE CANDLE AND PEERED IN. THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER, BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST. X CAN YOU SEE ANYTHING Q?

Jim Gillogly first reported the solution to Part 3 as a Triple Columnar Transposition. I was never able to duplicate that result, but eventually found a Complete Double Columnar Transposition ("Complete" meaning a full rectangle with no short or incomplete columns) with widths of 21 and 28 columns, with the columns taken off in order from right to left.

It was much later (December of 2002) that I came across this posting from Jim in sci.crypt:

I didn't actually solve it that way: I solved it as a double columnar transposition which put a bunch of English strings together; I reassembled them as a third irregular transposition. It's left as an exercise to the reader to see why they're equivalent... meaning that I never got around to working it out.

I think it's likely that the 21 and 28 columns probably put the solution to this double columnar beyond the capabilities of Jim's hill-climbing program, thus the reason for his original partial solution containing broken strings of English text.

Part 3 Gets a Bit Simpler

There was one thing about my double columnar solution that bothered me. Why take the columns off in such a simple order (right to left)? That would seem to negate a good portion of the security of the double columnar transposition.

Then Ferdinando Stehle sent me a message that shed some light on the issue. He had found a much simpler transposition. See the cut and paste from his message below:

The solution of the third part was a "simple" transposition:

*let's say $P(x)$ the letter of plaintext at position X
 $C(y)$ the letter of the ciphertext at position Y
the transposition is: $C(y) = P((y*191 + 192) \bmod 337)$*

That was back in mid 2001, so my memory is a bit foggy, but I think I might have had to swap the 191 and the 192 to make the formula work. But it did work. And it did show that the

transposition was considerably simpler than a double columnar. It also made some sense. In a 336 character long message, my double columnar, with 21 and 28 columns, would have had column lengths of 16 and 12 respectively. $16 \times 12 = 192$.

... and Simpler

It wasn't until recently (May 2003), when I dug out my Kryptos notes for another look, that I realized the significance of that formula. I, and I assume most others, had been working with a message length of 336. Both the double columnar and Stehle's formula worked great with a message length of 336. And 336 was such a "nice" number to work with. Plenty of factors. Lots of different size rectangles for columnar transpositions. It was just a nice number. I was happy to treat the "?" at the end as simply a divider between the messages.

But, add that "?" to the message and the length becomes 337... and the decryption becomes trivial. Count in 192. That's the first character of the plain text. Count another 192, looping back to the beginning. There's the second character. Continue and you'll uncover the entire message, ending with the question mark. It was so simple that a child could do it.

There are still things about Part 3 that bother me. Why the "Q" just before the question mark? Some have speculated that it was indicated "question mark." Not likely, since there was already a question mark.

A more plausible reason was to change the message length. I haven't done the math, but it would seem that this new simpler encryption/decryption scheme only works if the message length and the key number length have no common factors. (Otherwise you'd end up landing on the same letters more than once, while missing others completely.) The easiest way to avoid common factors is to make the message length prime – as in 337. So I speculate that the Q is simply a null character. The irony is that if "desparatly" had been spelled correctly there would have been no need for the Q. This is one question I'm dying to ask Sanborn after Kryptos is solved.

ID BY ROWS?

It was widely believed that the "ID BY ROWS" in Part 2 was a clue as to the method of encryption employed in Part 3. That is likely what inspired Jim Gillogly to approach it as a columnar transposition.

Elonka Dunnin, after learning of the simplified approaches outlined above, decided to look for an equally simple solution that was consistent with the "ID BY ROWS" concept. Her entire work can be found posted in the Kryptos group files section.

I think, in an effort to make it as simple as possible to extract the correct text, she may be over complicating the task. I think her very first step already makes the task elegantly simple and may very well be the solution Sanborn and Scheidt had in mind.

The first step requires the text to be arranged in equal length rows. (Because of the question mark, the last row will actually have 1 extra character.) The math works out for any equal length rows, but the solution is "prettier" with rows of 24 or 48. Here I will use 48, to be consistent with Elonka's layout.

ENDYAHROHNLSRHEOCPTIOIBIDYSHNAIACHTNREYULDSLLSLL
NOHSNOSMRWXMNETPRNGATIHNRARPESLNNELEBLPIIACAEMWT
WNDITEENRAHCTENEUDRETNHAEOTFOLSEDTIWENHAEIOYTEY
QHEENCTAYCREIFTBRS PAMHHEWENATAMATEGYEERLBTEEFOAS
FIOTUETUAEOTOARMAEERTNRTIBSEDDNIAAHTTMSTEWPIEROA
GRIEWFEBAECTDDHILCEIHSITEGOEASDDRYDLORITRKLMLH
AGTDHARDPNEOHMGFMFEUHEECDMRIPFEIMEHNLSSTTRTVDOHW?

As Elonka points out with the letters in red, a pattern quickly emerges. Starting at the red S in the last column simply count down 4, shifting one column to the left when you reach the bottom of a column. Following this pattern, the entire message can be recovered.

It is not necessary to know where the message starts. Beginning any place, and taking every 4th character quickly begins to deliver intelligible text. Trying 2 then 3, etc. would eventually result in a solution. In rows of 48 that solution comes quite quickly with a count of 4. Other row lengths may require much longer counts.

There is a slightly quicker solution, which will work with any arrangement of equal length rows. If you had any reason to believe that there would be a null Q at the end of the message, all that would be required is to start at the question mark and count to the Q. Since are starting at the end of the message and working backwards, you would count UP instead of down, and shift through the columns from left to right.

Since the question mark is on a column all it's own, the first count up actually moves you from the question mark to the bottom of the first row. You reach the Q on four. Continuing the count brings out the entire message, in reverse – QGNIHTYNA...

Is there any way you could have known about the null Q at the end? Unlikely, but not impossible. Consider the target audience – CIA employees. It is reasonable to assume that Scheidt, in his position at the CIA, would have worked out examples of ciphers with fellow employees and/or “students.” If he happened to regularly use a Q when a null was required there could have be a number of people that might expect the same to be true in Kryptos.

Another possibility is that no one was expected to anticipate *this* Q. The solution already seems simple enough. Additional aids were not necessary. It is possible that Scheidt intended this Q to be found, and to then serve as a hint for Part 4. After solving the substitution in Part 4, and finding an unusually large number of Qs, one could conclude that Q served as a null in Part 4 as well. There may even be a string of nulls, which could serve as a crib in the solution of the transposition portion of Part 4.

Implications on Part 4

The possible implications of such a simple solution to Part 3 should be obvious. Could Part 4 actually be much simpler than we think? I would argue yes. For some evidence of this see the next document in this series – *Kryptos, Part 4 – Substitution*.

Post any comments or questions in the Kryptos Group (<http://groups.yahoo.com/group/kryptos/>) or email me at gwarzin@yahoo.com.